Angle-Scanned Phase Illumination for Fixed-Focus 3D Fluorescence Imaging with Structural Reliability Analysis

Anonymous CVPR submission

Abstract

Fixed-focus 3D fluorescence imaging usually encodes depth in the detection optics. We instead encode it in the excitation, by scanning only the input angle of a plane wave through one fixed, defocused phase-only plate. Tilting the input shears the plate's three-dimensional speckle about the plate's conjugate plane, outside the specimen, so each depth receives its own sequence of excitation values, and a conventional low-NA camera at a fixed focus records one image per angle. A voxel-integrated measurement operator, assembled exactly from a fine-grid vectorial simulation, reduces the forward discrepancy of coarse-grid simulation from 25.7% to 1.07%, and standard regularised solvers (box-constrained total variation, multi-slice plug-and-play) recover the volume. In simulations on mouse neuron volumes from six cells, at equal deposited excitation dose with Poisson noise, the plate gives 0.19–0.43 µm depth error against 2.17–2.33 µm for an in-focus amplitude-pattern encoder and gains 2.4–3.8 dB PSNR on every crop over a no-plate angle scan. A defocused amplitude mask comes close: over three realisations of each, the plate leads by 0.3–1.2 dB PSNR and 0.02–0.08 foreground SSIM but not in depth error, so most of the depth encoding comes from defocus under the angle scan; halving the defocus costs up to 1 dB. We then ask which reconstructed connections the measurements establish. On a validated reduced field at the operating noise level, 48 of 72 connectivity queries admit both a connected and a disconnected volume consistent with the images, even where the reconstruction is correct; using all 256 directions at the same photon budget resolves none, a calibrated regularity prior does not remove the ambiguity, and sub-voxel model mismatch widens it.

1. Introduction

Three-dimensional fluorescence imaging normally records a stack of focal planes, moving either the specimen or the focus. Fixed-focus alternatives encode depth into planar images and recover the volume computationally. Most encode depth on the detection side, giving each depth a distinct point-spread function with a lenslet array, diffuser or phase mask [2, 7, 12, 49, 50]; others encode it in the excitation with speckle or patterned illumination, usually by translating or regenerating the pattern [13, 37, 51]. Each choice fixes what has to move or be replaced between measurements and how much depth information a planar image carries.

We study an excitation-side encoder in which nothing moves except the direction of the incident beam. A plane wave is scanned in angle across one fixed phase-only plate placed a short distance before an intermediate illumination image plane; a high-NA relay maps the resulting field into the specimen; and a conventional low-NA camera at a fixed focus records one image per angle (Fig. 1). The plate converts each input direction into a three-dimensional excitation pattern, and because a tilt translates the diffracted pattern by an amount that grows with propagation distance, different depths see different sequences of excitation across the scan. This gives depth-distinguishing measurements without axial scanning, refocusing, a moving diffuser or a reconfigurable modulator.

Turning these measurements into volumes needs a forward model that is accurate at the scale of the illumination structure. We simulate the excitation with a vectorial high-NA model on a fine grid and integrate it exactly over each reconstruction voxel. An illumination–emission factorisation that follows from linear incoherent fluorescence makes this tractable, and the full operator builds in about two minutes on one GPU. We reconstruct with two standard solvers, box-constrained total-variation least squares and multi-slice plug-and-play ADMM with one pretrained 2D denoiser on three orthogonal slice sets [29, 44, 53]; the encoder comparison holds with both.

A second question arises once volumes look convincing. A reconstruction may show two locations joined by fluorescent structure even when the measurements also admit a volume in which they are disjoint, so the prior, not the data, made the choice. We therefore complement the imaging study with a measurement-level analysis of connectivity: for a query, we either certify that every volume consistent with the images and a calibrated noise model agrees, or exhibit two consistent volumes with opposite connectivity.

Contributions.

1. **Acquisition.** A fixed-focus 3D fluorescence acquisition in which only the illumination angle changes through one fixed, defocused phase encoder. We explain its depth encoding by the depth-dependent shear of the excitation with input tilt, measure that shear numerically, and compare it under matched photon budgets with a no-plate angle scan, an in-focus amplitude-pattern (DMD) encoder and a defocused amplitude mask, all through the same vectorial relay, detector and solvers, over several encoder realisations and plate defocus distances.
2. **Forward model and reconstruction.** An exactly voxel-integrated operator built from a fine-grid vectorial model through an illumination–emission factorisation, which reduces the forward discrepancy of direct coarse-grid simulation from 25.7% to 1.07%, combined with standard regularised recovery. On mouse neuron volumes from six cells of three animals, dominant dendrites and their depth order are retained at eight-fold compression (4,608 measurements for 36,864 voxels).
3. **Structural reliability.** A query-level analysis of which reconstructed connections the measurements establish. On a validated reduced field at the system's operating noise level, many connectivity queries are not determined by the data, even when the reconstruction is correct. Using every available direction at a fixed photon budget resolves none of them, a calibrated regularity prior does not remove the ambiguity, and fine-grid model mismatch widens it.

2. Related work

Fixed-focus 3D fluorescence by detection coding. Light-field microscopy, DiffuserCam, Miniscope3D, Fourier light-field and related lensless designs give each depth a distinct detection point-spread function and recover a volume from one or few images with uniform excitation [1, 2, 7, 12, 27, 38, 49, 50]. Our camera is a conventional fixed-focus, low-NA detector; depth is encoded only in the excitation.

Excitation coding with speckle and structured light. Speckle and structured illumination are established for sectioning, super-resolution and 3D fluorescence [13, 14, 23, 25, 30, 33, 34, 45, 51]; most 3D variants still scan axially [30, 34, 51]. Pascucci *et al*. recover 3D fluorescence from a lateral scan of SLM speckle using its axial decorrelation, with point detection [37], as does multimode-fibre speckle [28]; DEEP and DEEP^2 image through scattering with patterned excitation [48, 54]. Tilted projection of a known pattern shifts it by depth-dependent amounts and separates depth sections without axial scanning [20]; multi-angle TIRF uses angle-dependent penetration [5].

Angle scanning through a fixed scatterer. The angular memory effect of thin scattering layers [9, 10, 36] has been used to excite fluorescence by scanning the incidence angle on a fixed diffuser, to recover a hidden 2D object from total fluorescence [3] or to improve 2D resolution with one camera frame per angle [52]; detection-side memory-effect methods image through scatterers [17, 21]. These works establish the physics we use. We combine one fixed, deliberately defocused random phase plate, a large 2D angle scan ($|\mathbf{s}| \leq 0.87$) relayed at NA 0.8, fixed-focus camera detection, a physically modelled operator and compressive 3D reconstruction; to our knowledge this combination has not been reported, though its ingredients are not new.

Forward models and reconstruction. The illumination–PSF product is the standard incoherent structured-illumination model [14, 33], Richards–Wolf gives the vectorial focus [41], and integrating responses over voxels parallels wave-optics light-field operators [7]; our factorised assembly is an efficient implementation of this model. Plug-and-play priors [44] and multi-slice fusion [29] are applied without modification. Unlike learned illumination or diffractive encoders [16, 22, 26], ours is one untrained random plate.

What the data determine. Set-membership estimation defines the parameters consistent with bounded-error data [32, 47]; Haldar bounds each entry over nearly data-consistent solutions [15], the per-voxel part of our analysis for an ℓ_2 ball. BUQO tests whether a structure in a MAP image is supported, using Bayesian credible regions [8, 24, 40]. Hallucination analyses study null spaces or bound what any method must hallucinate [4, 11, 19], and connectivity priors and integer programs segment curvilinear structure [35, 39, 43, 46]. We use standard encodings to characterise one instrument, not as a general method.

3. Angle-scanned phase-encoded 3D imaging

Acquisition. Let $\mathbf{r} = (x, y)$ and t index the illumination direction. A plane wave $u_t^{\mathrm{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\mathrm{ex}}\mathbf{s}_t\cdot\mathbf{r}}$ with transverse direction $\mathbf{s}_t = \cos\theta_t(\cos\alpha_t, \sin\alpha_t)$ passes a fixed phase-only plate $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$ (Fig. 1). We scan 16 azimuths and 16 signed elevations, $0.5 \leq \|\mathbf{s}_t\| \leq \sqrt{3}/2$, so the two scan coordinates set the direction and magnitude of a linear phase gradient across the plate. Angular-spectrum propagation carries τu_t^{in} over the plate-to-image-plane distance D; the objective pupil band-limits it to $\mathrm{NA} = 0.8$, and a vectorial Richards–Wolf relay [31, 41] gives the excitation intensity $I_t(\mathbf{r}, z) = \sum_{c\in\{x,y,z\}} |E_{t,c}(\mathbf{r}, z)|^2$ in the specimen. The camera records one planar image per direction; acquisition is therefore fixed-focus but sequential (256 frames for the full scan), not single-shot.

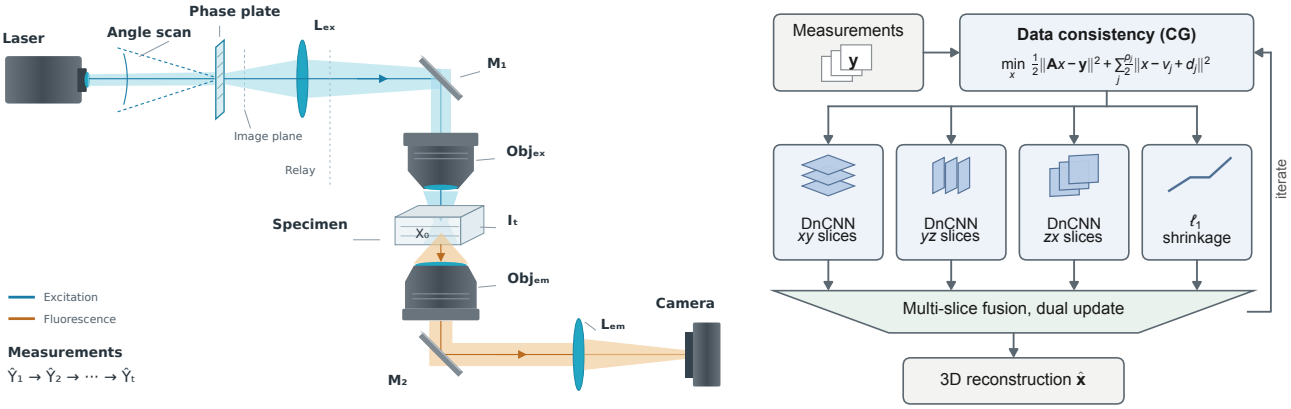


Figure 1. **Fixed-focus acquisition and reconstruction.** Left: a tilted plane wave illuminates a fixed phase plate placed 6.6 μm (specimen-space units) before an intermediate illumination image plane; an NA-0.8 vectorial relay maps the field into the specimen; a camera at a fixed detection focus records one image per illumination angle. Only the input angle changes: plate, specimen and focus are stationary, and nothing is scanned axially. Blue: excitation; orange: fluorescence. Right: multi-slice plug-and-play recovery alternates a conjugate-gradient data-consistency step with one pretrained 2D denoiser applied to the xy, yz and zx slices of a shared volume and an ℓ_1 shrinkage, fused by ADMM.

Why the angle encodes depth. Three properties of this arrangement matter. (i) *Without the plate*, a tilted plane wave that overfills the field has laterally uniform intensity at every depth, so changing the angle changes nothing in the specimen except, beyond the pupil, whether light arrives at all: the scan alone carries no depth information. (ii) *With the plate in an image plane*, a phase-only screen would give uniform intensity at focus and patterns symmetric about it. Placing the plate a distance D *before* the image plane makes the excitation a 3D speckle-like intensity whose planes differ and whose structure is not mirror-symmetric about focus. (iii) *Tilting the input* multiplies the field after a thin screen by the same linear phase, so the diffracted pattern at a further propagation distance $D + z$ is translated by approximately $(D + z)\,\Delta\mathbf{s}/\sqrt{1 - s^2}$ (the angular memory effect of a thin screen together with the Fresnel shift theorem). Different depths therefore translate at different rates as the angle changes, and every depth receives its own sequence of excitation values. We measure this shear in Sec. 6.1; it holds at 0.7–0.9 of this rate, with partial decorrelation between angles. By contrast, an amplitude pattern imaged in focus (as with a digital micromirror device) is real-valued at the focal plane, so its defocused intensities are equal at $\pm\Delta z$ and mirror depths are confounded.

Image formation. With fluorescence X_0, emission PSF emPSF and pixel region D_p, linear unsaturated fluorescence gives

$$Y_t = \mathrm{emPSF} * _3 (X_0 I_t), \qquad \hat{Y}^0_{t,p} = \frac{1}{|D_p|} \int_{D_p} Y_t(\mathbf{r}, z_{\mathrm{focal}})\, \mathrm{d}\mathbf{r}. \tag{1}$$

The product $X_0 I_t$ mixes specimen and illumination spatial frequencies before the low-NA blur and pixel averaging, so the detector need not resolve the illumination pattern; depth separation comes from how the excitation sequence varies across depths and angles.

Voxel-integrated operator. Optical simulation needs a fine grid ($0.08\,\mu$m optical, $0.32\,\mu$m analysis), reconstruction a coarse one ($0.64\,\mu$m). We keep the fine-grid measurement process B and let each reconstruction voxel represent constant fluorescence over a $2 \times 2 \times 2$ analysis patch, with replication U and averaging R:

$$\mathsf{A} = \mathsf{BU}, \qquad \mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f, \qquad \mathbf{y} = \mathsf{B}\mathbf{x}_f + \boldsymbol{\varepsilon}. \tag{2}$$

Measurements are always generated from the finer object $\mathbf{x}_f$, so the residual $\boldsymbol{\delta}_{\mathrm{disc}} = \mathsf{B}(\mathbf{x}_f - \mathsf{UR}\mathbf{x}_f)$ that the basis cannot represent is present in every experiment. Building A column by column would need one volumetric simulation per voxel and pattern. Because fluorescence is linear and incoherent, the response to an impulse at fine position $\mathbf{r}_i$ factorises into the excitation there and a pattern-independent emission footprint, $(\mathsf{B}_t \delta_i)_p = I_t(\mathbf{r}_i)\, g_p(\mathbf{r}_i)$, where g_p is the pooled detector image of an emitter at $\mathbf{r}_i$ (computed once), so that

$$\mathsf{A}_{(t,p),j} = \textstyle\sum_{i \in \mathcal{P}_j} I_t(\mathbf{r}_i)\, g_p(\mathbf{r}_i). \tag{3}$$

The factorisation itself follows from linear fluorescence physics; our contribution is to use it to integrate a fine-grid vectorial model exactly over the reconstruction basis at a cost of $O(TPF)$ for T patterns, P detector samples and F fine voxels, instead of $O(TNF \log F)$ for column simulation. The full $36{,}864 \times 36{,}864$ operator builds in about 110 s on

one GPU and matches the live forward model to 8×10^{-7} relative error (tolerance 10^{-5}).

Reconstruction. We combine measurement consistency with a pretrained 2D DnCNN [53] applied to the xy, yz and zx slice sets of one shared volume, $\mathcal{D}_o = \mathsf{P}_o^{-1}\mathcal{D}\mathsf{P}_o$, following plug-and-play priors and multi-slice fusion [29, 44]. With one auxiliary volume $\mathbf{v}_j$, dual $\mathbf{d}_j$ and penalty ρ_j per branch $j \in \{xy, yz, zx, \mathrm{sp}\}$, ADMM alternates

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \tfrac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \textstyle\sum_j \frac{\rho_j}{2}\|\mathbf{x} - \mathbf{v}_j^k + \mathbf{d}_j^k\|_2^2, \tag{4}$$

$$\mathbf{v}_o^{k+1} = \mathcal{D}_o(\mathbf{x}^{k+1} + \mathbf{d}_o^k), \quad \mathbf{v}_{\mathrm{sp}}^{k+1} = \mathcal{S}_{\lambda/\rho_{\mathrm{sp}}}(\mathbf{x}^{k+1} + \mathbf{d}_{\mathrm{sp}}^k),$$

with $\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + \mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}$, a conjugate-gradient solve of Eq. (4), soft thresholding $\mathcal{S}$, back-projection initialisation and a final clip to $[0, 1]$. As a non-learned solver we also compute $\arg\min_{0\le\mathbf{x}\le 1} \frac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda\,\mathrm{TV}(\mathbf{x})$ with isotropic 3D total variation [42] (ADMM [6], 300 iterations). No network is trained for the task; neither solver is a contribution.

4. What the measurements establish

A reconstruction is one volume that explains the images under a prior. For relational questions, such as whether a dendrite reaches a soma, we also ask whether *every* volume that explains the images agrees.

Consistency set and connectivity query. Given measurements $\mathbf{y} \in \mathbb{R}^M$ from k directions and per-row intervals $[\ell_i, h_i]$ that contain the noise-free measurement with stated probability, the measurement-consistency set is

$$\mathcal{C}(\mathbf{y}) = \{\mathbf{x} \in [0,1]^N : \ \ell_i \le (\mathsf{A}\mathbf{x})_i \le h_i, \ i = 1, \dots, M\}, \tag{5}$$

where the box is the reconstruction's intensity normalisation. For calibrated Gaussian noise we use the Bonferroni box $y_i \pm z\sigma$ with $z = \Phi^{-1}(1 - \alpha/2M)$, $\alpha = 0.05$, so that $\mathbf{x}_{\mathrm{gt}} \in \mathcal{C}(\mathbf{y})$ with probability at least 0.95 under the noise model. In a query region with 6-adjacency, a voxel is active if $x_v \ge \tau$, and p, q are connected if an active path joins them; equivalently the bottleneck $b(\mathbf{x}) = \max_{P:p\to q} \min_{v\in P} x_v \ge \tau$. The query is *certified connected* if $\min_{\mathcal{C}} b \ge \tau$, *certified disconnected* if $\max_{\mathcal{C}} b < \tau$, and *ambiguous* otherwise. Ambiguity is proven by exhibiting two volumes in $\mathcal{C}(\mathbf{y})$ with opposite connectivity, both re-checked independently of the optimiser (unscaled residual, breadth-first search and union–find). Problems that a solver cannot decide within its budget are reported as undecided, never as ambiguous.

Computation. Per-voxel bounds $L_v = \min_{\mathcal{C}} x_v$, $U_v = \max_{\mathcal{C}} x_v$ (two linear programs per voxel) certify the query when the voxels with $L_v \ge \tau$ already connect p, q, or those with $U_v \ge \tau$ do not. Otherwise mixed-integer programs with separator and flow encodings search for the two witnesses, and a joint verifier bounds $\min b$ and $\max b$ exactly. These encodings are standard; formulations, tolerances and validation are in the supplement. A certificate is a statement about all volumes allowed by the stated noise and object model, computed in floating point with independent re-checks; it is not a formal proof, and it does not hold for objects or errors outside the model.

5. Experimental setup

All experiments are simulations with the optics of Sec. 3: 532 nm excitation and detection, $n = 1$, excitation NA 0.8, detection NA 0.1, one random phase plate ($0.32\,\mu$m features, phases uniform on $[0, 2\pi)$, $D = 6.6\,\mu$m), 256 directions, $0.08\,\mu$m optical and $0.32\,\mu$m analysis grids, $0.64\,\mu$m reconstruction voxels, linear fluorescence in a homogeneous non-scattering medium. Two field sizes are used and reported separately.

Full-size imaging. Analysis grid $32 \times 96 \times 96$, reconstruction grid $16 \times 48 \times 48$ ($N = 36{,}864$, $30.72 \times 30.72 \times 10.24\,\mu\mathrm{m}^3$), 12×12 camera samples per image, so $M = 144k$ for k directions and compression $C = N/M \in \{1, 4, 8\}$ for $k = 256, 64, 32$ (five random direction subsets for $C > 1$). Reference volumes are two-photon stacks of mouse pyramidal neurons from the public DEEP^2 data [48], preprocessed as $32 \times 128 \times 128$ crops, min–max normalised, area-resampled laterally to the analysis grid and scaled to unit peak, with the $0.32\,\mu$m spacing assigned for simulation. A preliminary evaluation (supplement) used eight prespecified crops of *one* parent stack. For the encoder comparison we prespecified two crops from each of six distinct cells of three animals (cells sharing an animal are not fully independent), chosen from the reference alone by projected foreground area $\times$ depth spread; repeat time points of the same cells and stacks flagged for skipped frames were excluded. Measurements are always $\mathsf{B}\mathbf{x}_f$ from the fine object.

Photon budget for the encoder comparison. All four encoders share the relay, pupil, detector and voxel-integrated assembly; their excitation is simulated on an optical grid twice the field width (an overfilled illumination aperture, so beam edges lie outside the field), and the central plate, mask and DMD patterns are fixed in advance. Every encoder's excitation is normalised to the same mean over all directions and the whole analysis volume, i.e. equal excitation dose deposited in the specimen. Detected counts are $\mathrm{Poisson}(\kappa_{\mathrm{frame}} y_i)$ with $\kappa_{\mathrm{frame}} = \kappa_0\, 256/k$, a fixed total budget shared by the k frames. The exposure κ_0 of each specimen is set so that the proposed plate reaches a 15 dB op-

erating point at $k = 256$ (1000 photons per pixel per frame at the mean signal), from the forward model alone. *Primary:* every encoder receives this same exposure (equal deposited dose). *Sensitivity:* each encoder's exposure is rescaled so that it detects the same mean number of photons. We also report the fraction of incident power each encoder delivers through the pupil, which sets laser power but not specimen dose. All encoders use the same solvers, tuned per encoder and compression on two vascular crops of a separate stack (DnCNN prior, TV) or a held-out bead layout (sparsity prior), on grids that extend automatically at their edges ($C = 4, 8$; fixed grids at $C = 1$); no test volume is used for tuning. Plate and mask are evaluated in three realisations each, with seeds fixed in advance. Metrics: PSNR, foreground SSIM, local depth error, bead Jaccard index, axial localisation error and mirror-ghost energy.

Reduced structural protocol. Certification is exact optimisation with dense rows, so it is run on a reduced field of the same optics: analysis grid $8 \times 24 \times 24$, reconstruction grid $4 \times 12 \times 12$ ($N = 576$), 3×3 camera pixels per image, $M = 9k$. The operator agrees with the live simulator to 3×10^{-7}. Queries use the central $4 \times 4 \times 4$ region with 6-adjacency and $\tau = 0.4$; 24 prespecified synthetic phantoms cover filaments, staircases, alternative-depth routes, weak and missing bridges, bead chains, rings, a dim filament just above τ and dense random volumes; endpoints are the prespecified structure ends (oracle endpoints). The baseline uses $k = 32$ nested-grid directions; Gaussian noise at 15 and 25 dB with three seeds, with either a fixed total photon budget or fixed per-frame exposure; confidence boxes are Bonferroni at $\alpha = 0.05$ ($z = 4.30$), with row+SVD and Poisson (Garwood) sets as secondary checks. Solves use HiGHS [18] with 60 s per solve and a 600 s rerun of every undecided problem.

Structural experiments. *E1*: for 13 reduced problems (nine ambiguous, four certified controls) we reconstruct from the recorded measurements with box-constrained least squares, with and without a TV term whose weight is tuned on ten calibration phantoms, and compare with the certified outcome and witnesses. The full-size PnP solver is not used here: DnCNN's receptive field (about 35 pixels) exceeds the 4×12 slices of the reduced field. *E2*: the set is restricted to $\{\mathbf{x} \in \mathcal{C} : \mathrm{TV}(\mathbf{x}) \le R\}$ with an exact linear encoding, $R_{\mathrm{cal}} = 31.9$ (the largest TV of the calibration phantoms) and $2R_{\mathrm{cal}}$, $4R_{\mathrm{cal}}$. *E3*: measurements come from fine objects with sub-voxel structure (two models); an allowance $\epsilon_i = c(|y_i| + z\sigma)$ is calibrated on the calibration phantoms only (maximum or 99.9% row quantile of $|\delta_i|/(\mathsf{A}\mathsf{R}\mathbf{x}_f)_i$), frozen, and tested for coverage on held-out objects.

		multi-slice PnP			box-constrained TV			beads (PnP)	
C	encoder	PSNR	FG-SSIM	depth	PSNR	FG-SSIM	depth	depth	ghost
1	phase plate	35.73	0.814	0.357	38.15	0.881	0.187	0.00	0.033
	amp. mask	35.24	0.788	0.347	37.59	0.857	0.191	0.00	0.030
	no plate	33.06	0.655	0.378	34.39	0.682	0.295	0.21	0.085
	in-focus DMD	28.96	0.501	2.168	28.48	0.435	2.285	3.84	1.007
4	phase plate†	35.40	0.801	0.389	37.34	0.852	0.223	0.01	0.013
	amp. mask†	34.88	0.778	0.382	36.79	0.823	0.239	0.01	0.017
	no plate	32.49	0.618	0.501	34.74	0.703	0.302	0.32	0.160
	in-focus DMD	28.91	0.493	2.168	28.49	0.465	2.326	2.94	1.013
8	phase plate†	34.86	0.784	0.430	37.00	0.847	0.243	0.01	0.039
	amp. mask†	34.14	0.744	0.446	36.74	0.827	0.237	0.08	0.048
	no plate	32.42	0.623	0.521	34.34	0.676	0.342	0.49	0.211
	in-focus DMD	28.83	0.482	2.169	28.53	0.473	2.311	2.60	1.013

Table 1. **Encoder comparison** (mean over 12 test crops; depth errors in μm; beads: depth error of detected beads and mirror-ghost energy ratio). †PnP: mean over three plate or mask realisations; TV and all other rows: realisation 0. Equal deposited excitation dose; Poisson noise at a 15 dB operating point for the plate. Paired statistics in the text and supplement.

6. Results

6.1. Optical encoding under matched photon budgets

Figure 2 and Tab. 1 compare the phase plate with three encoders that share its relay, detector, dose, objects, solvers and tuning protocol. As predicted (Sec. 3), the plate's excitation shears with tilt at 0.7–0.9 of the thin-screen rate about a pivot that follows the plate: 3.2, 6.0–7.1 (three realisations) and 15 μm from focus for $D = 3.3$, 6.6 and 13.2 μm. Per-pixel excitation sequences decorrelate within about 1 μm of depth and are uncorrelated at mirror depths (median 0.05), whereas without the plate every depth sees the same sequence and the in-focus DMD gives identical sequences at mirror depths. Point emitters at $k = 32$ are assigned a wrong depth in 1.5–1.8% of cases with the three plates, 4.4–5.3% with three defocused amplitude masks, 25% with the DMD (half at the mirror depth) and 89% without the plate (94% for guessing).

The reconstructions follow the same order with both solvers. Against the DMD the plate reduces depth error from 2.17–2.33 to 0.19–0.43 μm and gains 6.0–9.7 dB PSNR on every crop; DMD beads appear at their mirror depths. Against the no-plate scan it gains 2.4–3.8 dB and 0.15–0.20 foreground SSIM on every crop (PnP, $C = 8$: 95% interval $[1.3, 3.8]$ dB), and no-plate bead errors reach 0.5 μm. The defocused amplitude mask comes closest; its first realisation proved the weakest of three at $C = 8$ (33.7 vs. 34.4 dB), so the 1.1 dB margin over it alone overstates the gap. Over three realisations of each, seeds fixed in advance, the plate's PnP PSNR varies by at most 0.12 dB; all nine plate–mask pairings favour the plate in PSNR ($+0.4$ to $+1.2$ dB at $C = 8$, $+0.4$ to $+0.6$ at $C = 4$) and foreground SSIM ($+0.02$ to $+0.08$), each matched pair with an interval above zero, but not in depth error (-0.04 to $+0.10$ μm). With TV the plate leads by 0.26 dB at $C = 8$ ($[-0.04, 0.55]$; equal depth error)

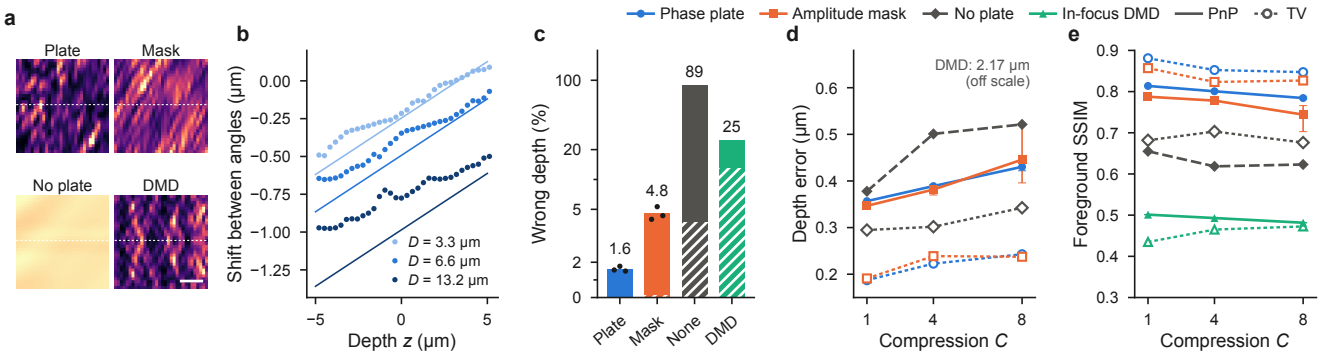


Figure 2. **Optical encoding under matched photon budgets** (full-size field, equal deposited excitation dose, overfilled illumination aperture). (a) One excitation pattern per encoder, xz section through the specimen depth ($\pm 5\,\mu$m, focus dotted), each tile scaled to its own 99.5th percentile. (b) Lateral shift of the plate's excitation between two scan angles at every depth for plate defocus $D = 3.3$, 6.6 and $13.2\,\mu$m; lines: thin-screen prediction, sheared about the plate's conjugate plane at distance D. (c) Point emitters assigned a wrong depth by maximum likelihood with unknown brightness at $k = 32$ (dots: three plate and three mask realisations; hatched: mirror depth). (d, e) Reconstructions of 12 neuron crops from 6 cells: local depth error and foreground SSIM versus compression, multi-slice PnP (solid; plate and mask at $C = 4, 8$: mean and range over three realisations) and box-constrained TV (dashed); the DMD's depth error of $2.17\,\mu$m is off scale.

and 0.56 dB at $C = 4$ and $C = 1$. Phase modulation thus adds a small, consistent gain in intensity fidelity, threefold fewer point-emitter depth errors and higher transmission (28% vs. 21% of the incident power, field-sized aperture); most depth information comes from defocus under the angle scan.

Defocus, tuning and solver. Halving the defocus to $3.3\,\mu$m costs 1.0 dB, 0.08 foreground SSIM and $0.15\,\mu$m depth error at $C = 8$ (12/12 crops; 0.4 dB at $C = 4$), as a smaller shear per angle step leaves neighbouring depths more alike; doubling it to $13.2\,\mu$m changes PSNR by -0.1 dB (interval includes zero). $D = 6.6\,\mu$m was fixed before this study. Widening every tuning grid, with automatic extension at its edges, changed one neuron selection (no plate, $C = 8$: $+0.43$ dB, gap to the plate 2.8 $\rightarrow$ 2.4 dB) and reproduced all other results to 10^{-3} dB. The non-learned TV solver, tuned by the same protocol, beats multi-slice PnP for every encoder but the DMD (plate, $C = 8$: 37.0 vs. 34.8 dB, 0.24 vs. $0.43\,\mu$m) without changing the ranking; its 300-iteration solves are not converged to 10^{-3} relative change, but 1000 iterations change PSNR by at most 0.2 dB (16 test solves at $C = 8$ and 1).

A simulation confound. When the simulated illuminated aperture equals the $30.7\,\mu$m field, its hard edge, $6.6\,\mu$m before the image plane, is itself a defocused amplitude encoder (edges sweep by $z \tan\theta$). With it, the no-plate scan and the mask matched or exceeded the plate in depth error (supplement). All results above use an aperture twice the field width; the DMD comparison holds in both geometries.

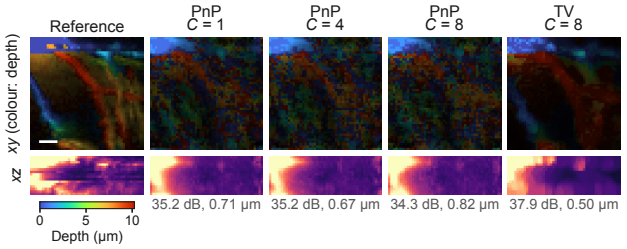


Figure 3. **Full-size recovery with the phase plate** on the median test crop (by plate PSNR at $C = 8$), one direction subset. Top: lateral maximum-intensity projection, colour coding the depth of the brightest voxel; bottom: xz projection. Identical display scale for all panels; below: PSNR and local depth error of this crop. Bar: $5\,\mu$m.

6.2. Volumetric recovery under compression

On the 12 test crops from six cells (overfilled aperture), the plate's reconstructions reach 35.7 ± 2.4 dB PSNR, 0.81 ± 0.11 foreground SSIM and $0.36 \pm 0.18\,\mu$m local depth error at $C = 1$, and 34.8 ± 2.1 dB, 0.78 ± 0.12 and $0.43 \pm 0.20\,\mu$m at $C = 8$, i.e. from 4,608 measurements for 36,864 voxels (mean $\pm$ s.d. over crops). Compression costs every crop: the paired changes from $C = 1$ to $C = 8$ are -0.92 ± 0.31 dB PSNR and $+0.074 \pm 0.024\,\mu$m depth error, worse on 12 of 12 crops. With TV the plate reaches 38.1 dB, 0.88 and $0.19\,\mu$m at $C = 1$ and 37.0 dB, 0.85 and $0.24\,\mu$m at $C = 8$. With PnP, beads are all detected at every compression (Jaccard 1.00) with depth errors of at most $0.02\,\mu$m and mirror-ghost ratios of 0.01–0.05. In Fig. 3 the bright shallow region and the main deep dendrite keep their depth order at every compression;

PnP leaves granular background and axial elongation, TV a smoother volume.

6.3. Forward-model accuracy and ablations

On a held-out neuron crop of the preliminary study, voxel integration reduces the noise-free relative forward discrepancy from 25.7% (direct simulation at the $0.64\,\mu\mathrm{m}$ reconstruction spacing) to 1.07%, and the reconstruction's depth error from 1.859 to $0.177\,\mu\mathrm{m}$ (PSNR 28.1 to 37.2 dB), because the NA-0.8 excitation needs sampling finer than the $0.64\,\mu\mathrm{m}$ grid. Three-orientation fusion improves PSNR over single orientations (37.22 vs. 36.26–36.81 dB) and sparsity alone (33.99 dB) but not depth error (0.177 vs. $0.163\,\mu\mathrm{m}$ for the best orientation).

6.4. Reconstruction appearance versus measurement-supported connectivity

Figure 4 contrasts what a reconstruction shows with what the measurements support, on 13 reduced problems (nine proven ambiguous, four certified controls; all cases in the supplement). All 26 reconstructions (two estimators) lie inside $\mathcal{C}(\mathbf{y})$: they are measurement-consistent volumes chosen by a prior, like the witnesses. The reconstruction reproduces the true topology in 11 of 13 cases, often with a clear margin (P01: bottleneck 0.64 vs. $\tau = 0.4$), yet in nine of the 13 the same images are explained by a volume of the opposite topology. In the dense random volume P24 the reconstruction reports a disconnection at both SNRs while the truth is connected, and it fits the data more closely than the truth does (largest normalised residual 0.24 vs. 0.83). Certified controls agree with every reconstruction, as they must. The opposite-topology volumes are not rough artefacts: for the filament phantoms their total variation is within 15% of the truth's, and they differ from it mainly by moving brightness between depth-adjacent voxels.

6.5. Reliability under noise, photons, priors and model mismatch

Noise and photons. With exact data, the reduced operator certifies every query once it is square and full rank ($k = 64$), uniqueness by linear algebra; its 8.9×10^5 condition number is why this does not survive noise. At 15 dB, 48 of 72 queries are proven ambiguous at $k = 32$ (the same 16 phantoms in every seed), and 12 at 25 dB (Tab. 2); an overfilled-aperture operator gives the same counts. On that field the no-plate control certifies 1 and 6 of 24 seed-0 queries at 15 and 25 dB against 8 and 20 for the plate. Spreading the same photons over all 256 directions resolves none of the 60 ambiguous queries, nor shrinks per-voxel uncertainty (supplement); eight times the photons resolve 6 of 48 and 3 of 12, all through per-voxel bounds. The comparison followed a rule fixed before aggregation, and all solves that hit 60 s were rerun at 600 s, leaving none undecided. Joint reasoning adds little: on 1464 earlier screening queries, per-voxel bounds decide 78% and the integer-programming verifier certifies 6 more (0.4%), none under the calibrated noise model.

setting (reduced field, $\tau = 0.4$)	certified	ambiguous	other
$k = 32$, 15 dB (72 queries)	24	48	0
$k = 256$, same photons	24	48	0
$k = 256$, $8\times$ photons	30	42	0
$k = 32$, 25 dB (72 queries)	60	12	0
E2: TV $\leq R$, truth feasible (18 runs)	0	18	0
E2: TV $\leq R$, truth infeasible (9 runs)	0	4	5 empty
E3: exact coarse model (16)	0	1	15 empty
E3: noise only (32)	0	4	28 empty
E3: noise + calibrated mismatch (64)	1*	63	0

Table 2. **Structural outcomes.** Top: 24 phantoms $\times$ 3 seeds. E2: the nine ambiguous E1 cases under a calibrated total-variation budget at $R_{\mathrm{cal}}, 2R_{\mathrm{cal}}, 4R_{\mathrm{cal}}$. E3: eight held-out phantoms with fine-grid measurements under two sub-voxel models. *This certificate contradicts the truth; the calibrated allowance failed to cover that object. Wrong certificates otherwise: 0.

A calibrated regularity prior (E2). Restricting $\mathcal{C}(\mathbf{y})$ to volumes with total variation at most $R_{\mathrm{cal}} = 31.9$, the largest value among ten calibration phantoms (or $2R_{\mathrm{cal}}$, $4R_{\mathrm{cal}}$), leaves every query ambiguous whenever the true object satisfies the budget (18 of 18 runs): new opposite-topology witnesses exist that are as regular as the truth (P20: 17.3 and 17.5 vs. 17.5). When the budget excludes the truth, the set becomes empty (5 runs), which flags the conflict, or stays ambiguous; no run produced a certificate.

Fine-grid model mismatch (E3). With sub-voxel structure in the objects, the coarse model with calibrated noise is inconsistent with the data in most cases ($\mathcal{C}(\mathbf{y})$ empty in 15 of 16 exact-model and 28 of 32 noise-only runs), which the method reports instead of a conclusion. An allowance $\epsilon_i = c(|y_i| + z\sigma)$ calibrated on separate phantoms covers the coarse truth for 83–92% of held-out objects, but then every query is ambiguous, including the two coarse-model controls, with a median 52.5 of 64 query voxels uncertain. All coverage failures come from one held-out phantom whose residual (up to 25%) exceeds the calibration range (13%), and in one of its runs a disconnection was certified although the structure is connected. Certificates are only as reliable as the coverage of the error model, which simulation can calibrate but not guarantee.

7. Discussion and limitations

What the optics provide. Depth information comes from a defocused encoder under an angle scan: its excitation shears with tilt about a plane outside the specimen, and the

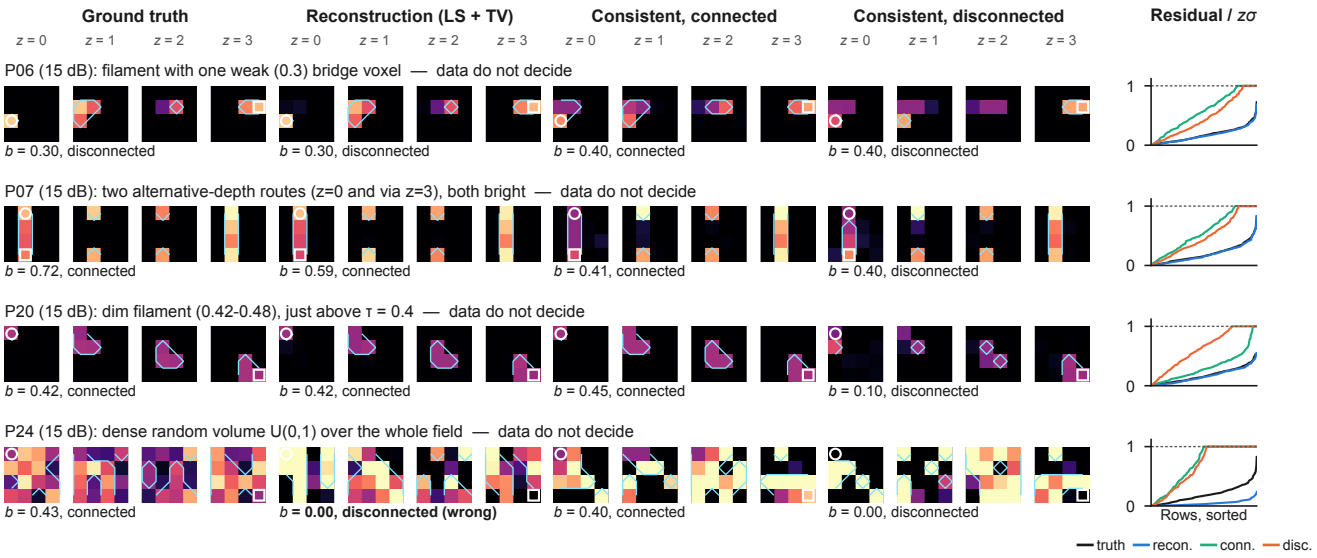


Figure 4. **One set of measurements, opposite topologies** (reduced field, 32 directions, 15 dB). Each row shows the $4 \times 4 \times 4$ query region as four depth slices: ground truth; box-constrained least squares + TV reconstructed from the recorded measurements; a volume in $\mathcal{C}(\mathbf{y})$ that connects p (circle) to q (square); a volume in $\mathcal{C}(\mathbf{y})$ that does not. Cyan outlines: voxels above $\tau = 0.4$; b: bottleneck. Right: sorted residuals $|(\mathbf{A}\mathbf{x} - \mathbf{y})_i|/z\sigma$; all curves stay at or below 1, so all four volumes explain the same images within the 95% box. The reconstruction is correct in the first three rows and wrong in the last; in all four the data do not decide the query.

defocus breaks the mirror symmetry that defeats in-focus amplitude patterns; a plain angle scan provides almost none of this. Phase versus amplitude modulation matters less: the plate is consistently but modestly better than a defocused amplitude mask, in intensity fidelity rather than depth, and wastes less light; sufficient defocus matters more. The solver matters too: a tuned TV solver beats our multi-slice PnP by about 2 dB, so the encoder ranking, not the absolute numbers, is the robust result. Encoder simulations need an overfilled aperture, since an aperture as small as the field is itself an amplitude encoder.

What the measurements do not establish. At the operating SNR, many connections that a reconstruction shows are not determined by the data: the prior chooses between consistent volumes, and in one dense case it chooses wrongly. More directions at a fixed photon budget do not change this; more photons change it slightly. Connections reported from such a system should therefore come with the outcome of the query, not a rendering alone, and photon budget, not angular sampling, is the lever when a particular connection matters.

Limitations. All results are simulations; there is no hardware validation. The optics assume linear fluorescence, a homogeneous non-scattering medium, equal excitation and emission wavelengths and an ideal relay; the reference volumes are two-photon stacks with an assigned simulation spacing. The encoder comparison uses three plate and three mask realisations but one set of DMD patterns, robustness runs at $C = 4$ and 8 only, TV for realisation 0 only, 12 crops from six cells of three animals (crops of a cell, and cells of an animal, are not independent) and shot noise only. The overfilled aperture was introduced after the first comparison revealed the aperture confound; both are reported. The structural analysis runs on a $4 \times 12 \times 12$ field with oracle endpoints, a fixed threshold and simulated noise; reconstruction and certification were never run jointly at full size, and full-volume certification is not claimed. Certificates are computed in floating point with independent re-checks, hold only for the stated noise and object model, and can be wrong when the model fails to cover the object, as one mismatch run shows; coverage is empirical, not a finite-sample guarantee.

8. Conclusion

Scanning only the input angle of a plane wave through one fixed, defocused phase plate gives a fixed-focus fluorescence microscope depth-distinguishing measurements, because the excitation shears with tilt. With an exactly voxel-integrated operator and standard solvers the volume is recovered without axial scanning, also from eight times fewer measurements than voxels; at equal dose the plate clearly beats in-focus amplitude patterns and a plain angle scan, and modestly a defocused amplitude mask. Many reconstructed connections, however, are not determined by the data at the operating noise level.

LLM Usage Statement

LLMs were used for grammar and fact checking.

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